

DATABASES INTRODUCTORY ELEMENTS (SEMINAR 1)

GENERAL OBJECTIVE OF THE COURSE



Students acquire the necessary skills for using the main concepts regarding database design and development in an economic environment.

ACQUIRED COMPETENCES



- Designing a database
- Formulating queries on a database
- Performing database updates
- Performing query optimizations

ABOUT ME – TEACHING ACTIVITY



COURSE & SEMINAR

- Associate Professor Anda Belciu (Velicanu), PhD – anda.belciu@ie.ase.ro (Office 2017 C)
- Databases (2nd year, sem 1) – English (Mondays, 10.30 – 2209 & Fridays, 12.00 - 2017)
- Software Packages (3rd year, sem 2) – English
- Databases for Business (BDSA) Master: Advanced Database Systems, Integrating Databases in Cloud Environment
- Applied Data Analytics (ADA) Master: Data Management for Data Science

EVALUATION

- Final exam – 50%
 - Written examination, on the computer
 - Multiple answer questions + Practical SQL exercises
- Seminar evaluation – 50%
 - Practical test 50%
 - Individual project 30%
 - Seminar activity 20%
- **Final exam entry conditions :**
 - A minimum of 5 for the seminar
and
 - Presenting the project
- **Discipline finalization conditions:**
 - a minimum grade of 5 for the seminar
 - a minimum grade of 5 for the final exam
 - If these 2 conditions have been met, the final grade of the discipline is calculated as the average between the 2 grades.



EVALUATION

- Seminar activity
 - Attendance
 - Active participation in discussions during classes
 - Exercises solved during classes
 - Homework

EVALUATION

- Individual project "Design and implementation of a database in the economic field, using Oracle "
 - The projects will be uploaded in the dedicated section on the **online.ase.ro** before the last seminar.
 - The presentation of the project is conditioned by its upload by the deadline.
 - The projects will be presented personally at the seminar, at the group of which the students are part.
 - Presenting the project is mandatory in order to be accepted for the final exam.
 - Project requirements – see online.ase.ro

EVALUATION

- Practical test

- will be held in week 12 *;
- It is mandatory that by the time of the test each student has created and populated the seminar DB schema on which he/she will solve the project requirements;
- topics from all the material discussed so far.

* Depending on the conditions related to the infrastructure and other organizational details, the test may be scheduled in another week. The final planning will be indicated at the seminar.

WORKING ENVIRONMENTS

- Oracle Database
 - Locally installed on personal computers
 - or
 - Connect to the faculty's Oracle external server – accounts next week
- Sql Developer –
 - <https://www.oracle.com/tools/downloads/sqldev-downloads.html>
 - or
- Oracle APEX
 - <https://apex.oracle.com/en/>
 - In order to work with it, it is necessary to create a free account on the platform.

RESOURCES

- <http://bd.ase.ro> & <https://online.ase.ro/>
- I.Lungu (coord.), A.Bâra. C.Bodea, I.Botha, V.Diaconița, A.Florea, A.Velicanu - Tratat de baze de date. Vol I. Baze de date. Organizare, proiectare și implementare, editura ASE, București, 2011, România
- A.Bâra, I.Botha, A.Fodor, I.Lungu, S.Oprea - SGBD Oracle. Limbajul SQL, editura ASE, București, 2016, România
- I.Lungu - Baze de date Oracle. Limbajul SQL, editura ASE, București, 2006, România
- M.Velicanu - Dicționar explicativ al sistemelor de baze de date, editura Economică, București, 2005, România
- **A. Belciu, Introduction to Databases, ASE Publishing House, 2016, Romania.**

OPTIONAL ACTIVITY



- Free optional online courses,
 - 1st Semester: Oracle Academy Database Design and Programming with SQL
 - 2nd Semester 2*: Oracle Academy Database Programming with PL/SQL
- Course graduation conditions:
 - completion of online courses (all quizzes and exams) by **16th of December 2024**
 - pass an additional online final exam held in week 12
 - maximum 3 absences course + seminar
 - at least a final grade 7 for the Database discipline
- Benefits:
 - course finalization diploma
 - discount vouchers for Oracle University courses or certification exams (www.oracle.ro)
 - bonuses

** PL/SQL Course can be accessed ONLY by the students that achieved the SQL diploma*

THEORETICAL INTRODUCTION

- DB
- TABLE (attributes + tuples)
- LINKS (1 to 1, 1 to many, many to many)
- KEY (primary/foreign, simple/complex)
- SQL
- Normal Forms
- Design of the DB structure

NF1

- All the table values contain elementary values (non-decomposable, not repetitive).
- Ex: last_name+first_name=name
address

NF2

- NF1+ each non-key attribute of the table is **fully functionally dependent** on the key.
- Ex: job_history(employee_id, job_id, start_date, end_date, employee_name)

NF3

- NF2 + each non-key attribute is **non-transitively dependent** on the key.
- Ex: jobs(job_id, job_title, dept_id, dept_name)

OPTIONAL NF

- NF4: NF3 + the table doesn't contain 2 or more multi-value dependencies.
- NF5: NF4 + each join dependency is generated through a key candidate for the table

DESIGN OF THE DB STRUCTURE

- Conceptual Design = establishing the data collections (tables), the fields they contain and the links between them
- Logical (external) Design = is made using views and other mechanisms that grant rights to users for accessing the DB
- Physical (internal) Design = physical data and organizing the way it is accessed (using indexes and clusters)

HOMEWORK

- Install SQLDeveloper
- Design your own database schema on a piece of paper